An Experimental Parametric Investigation of RMP Geometry and Calibration

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Introduction

- Remote Microphone Probes (RMPs) measure airfoil surface pressure fluctuations
- The RMP geometry attenuates and offsets pressure signals
  - Calibration is required to obtain a transfer function (TF)[1]
- TFs are sensitive to numerous variables
- Current methods in literature:
  - High variability
  - Limited frequency range

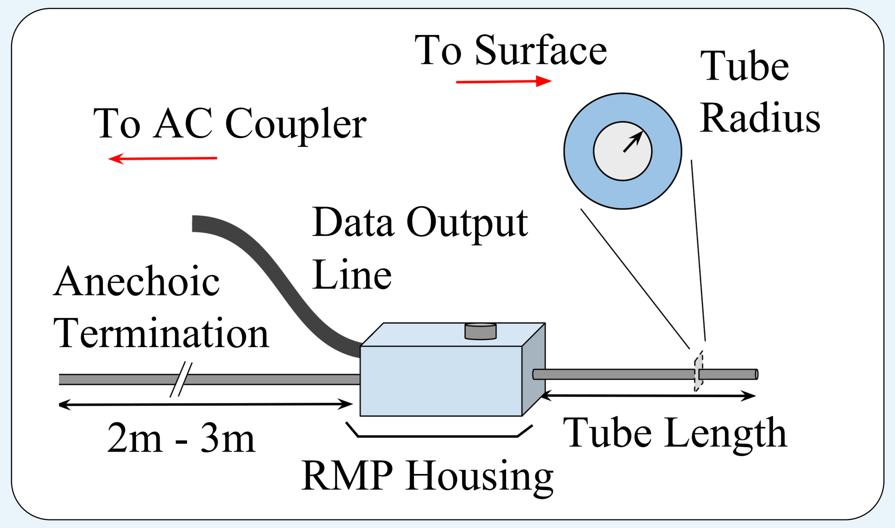


Figure 1: The RMP geometry used in this study, along with the governing variables.

Objective: The effect of 16 variables on TF behaviour were investigated to provide a basis for improved calibration.

Methods

- TFs were compared through: Coherence, Attenuation difference, Tortuosity[2], and Repeatability
- Two RMP apparatuses used:
  - Aluminum surface
  - Resin surface

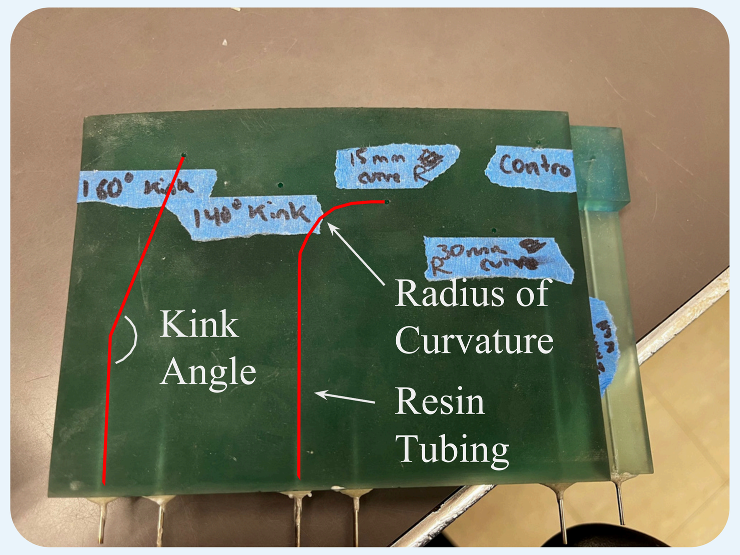


Figure 2: Resin geometry used in this study, along with the governing variables.

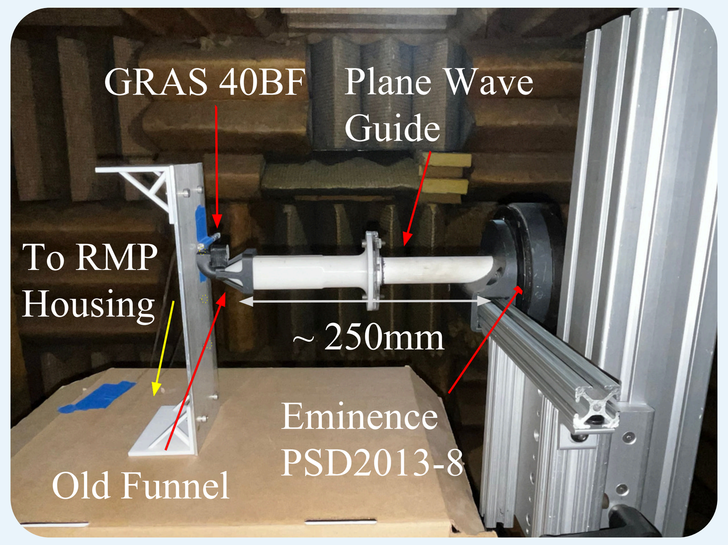
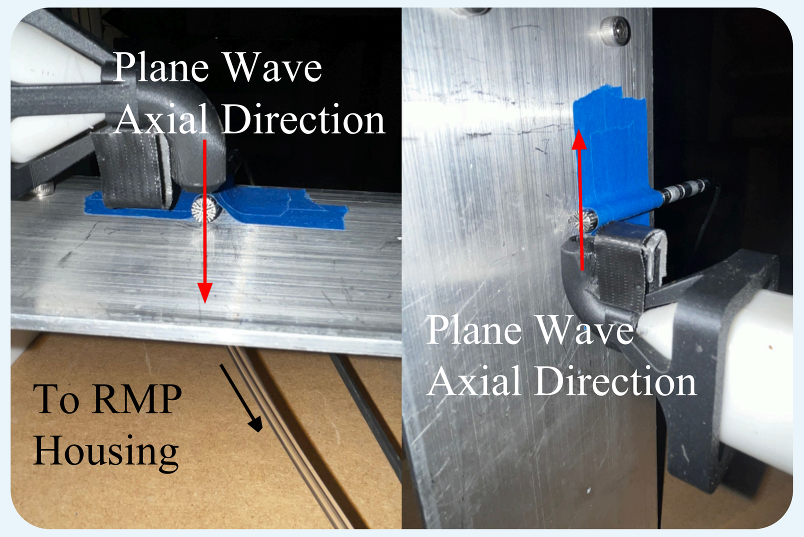


Figure 3: The experimental apparatus, with the aluminum configuration is pictured.

- Two speaker orientations were investigated



a) Perpendicular b) Parallel

Figure 4: Two speaker orientations tested in this study.

Results

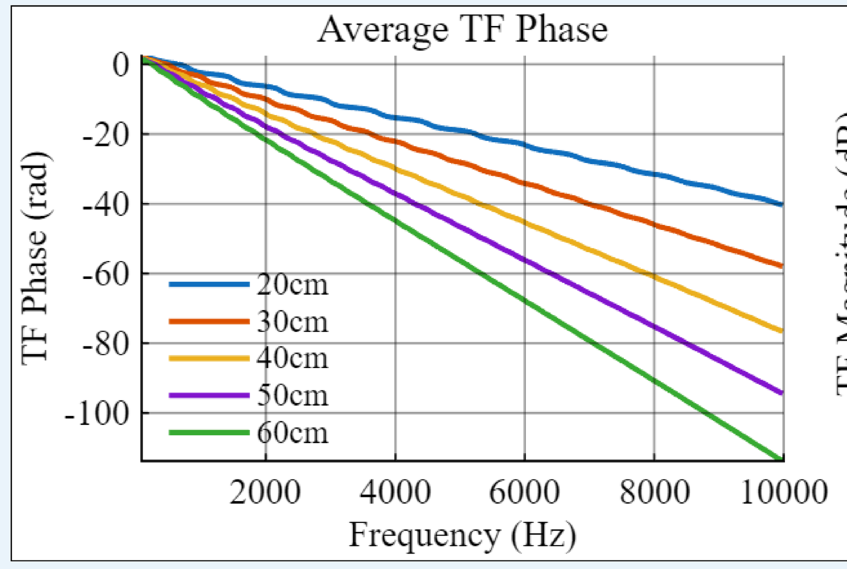


Figure 5: Effect of tube length on transfer function phase.

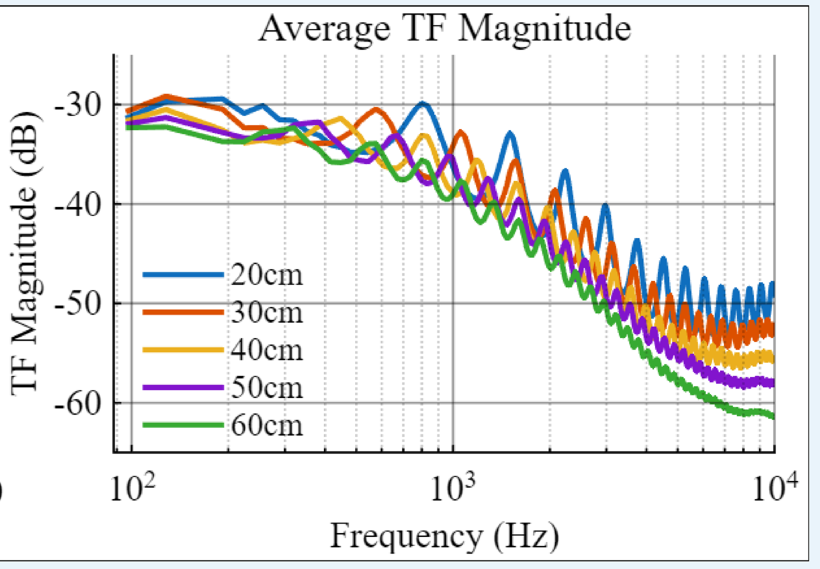


Figure 6: Effect of tube length on transfer function magnitude.

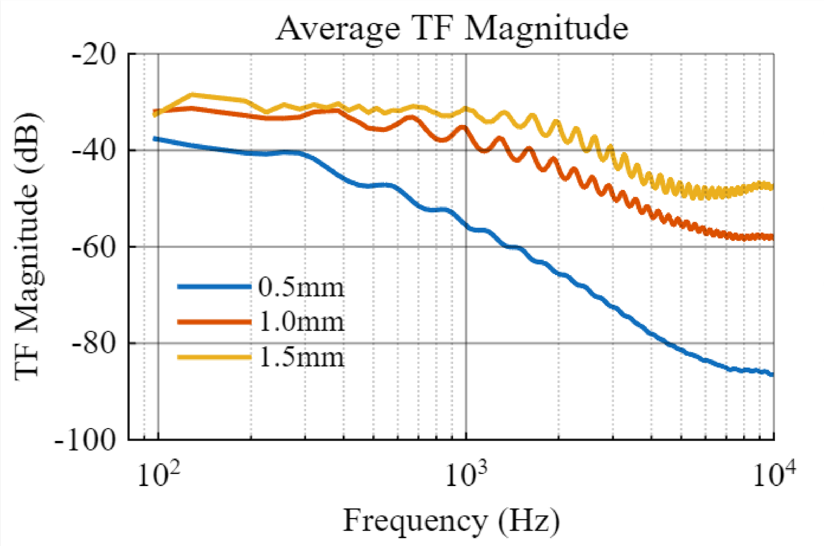


Figure 7: Effect of tube radius on transfer function magnitude.

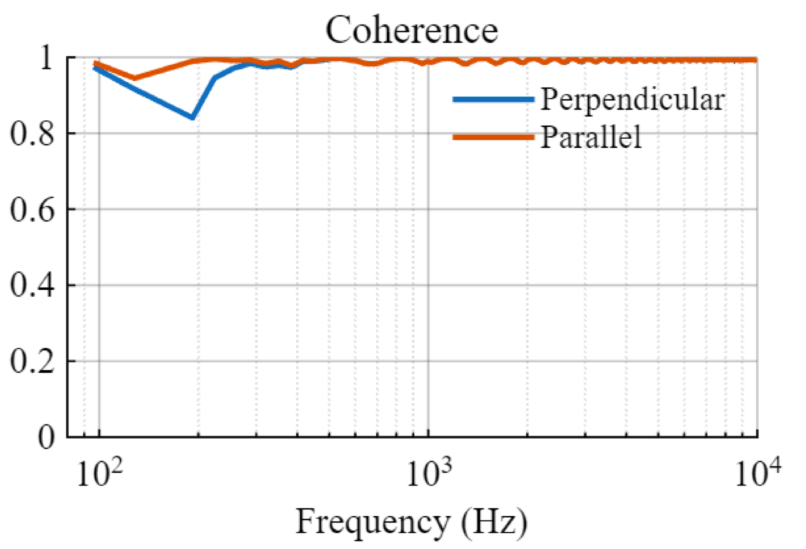


Figure 8: Coherence of different speaker orientations

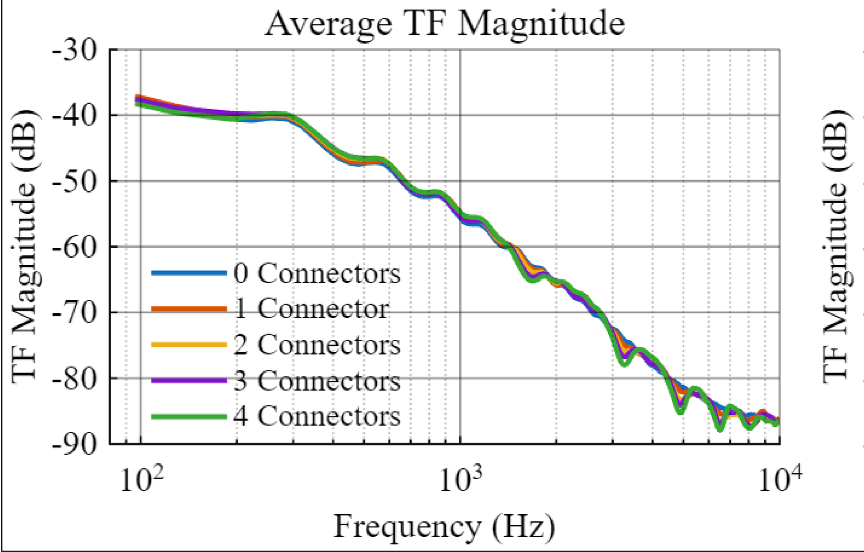


Figure 9: Effect of the number of hypodermic connectors on transfer function magnitude. Connectors were spaced periodically in 10cm increments.

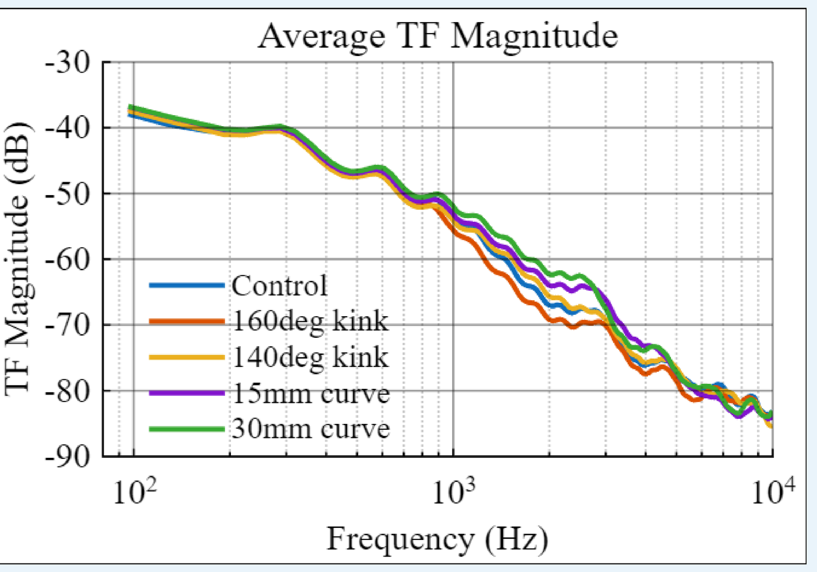
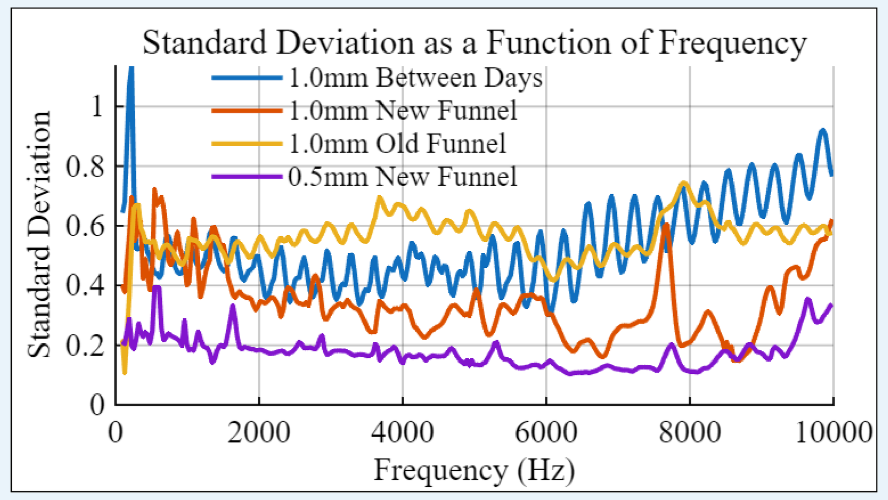


Figure 10: Effect of resin tube geometries on transfer function magnitude.

Figure 11: All SDs were calculated using five trials, with the exception of the 0.5mm radius plot which was calculated using four.

- 0.56 → 0.18



Radius (mm)	Attenuation Difference (dB)	Total Tortuosity (min. 1.0)
0.5	48.9	1.12
1.0	27.1	2.14
1.5	21.6	3.68

Table 1: Tube radius, net attenuation and tortuosity. Total tortuosity is the average of phase and magnitude.

Length (cm)	Attenuation Difference (dB)	Total Tortuosity (min. 1.0)
20	23.8	3.37
30	25.3	2.97
40	26.1	2.53
50	27.1	2.14
60	29.3	1.78

Table 2: Tube length, net attenuation and tortuosity.

Discussion

- Radius controls attenuation[3]; Length controls both phase slope and line resonances
- Attenuation and tortuosity have an inverse relationship

$$\alpha = i^{3/2} R \sqrt{\frac{\omega}{\nu}},$$
$$f_L = \frac{nc}{2L},\ k_L = \frac{\omega L}{c},$$

Improvement: Decrease length and radius

- Abrupt changes in radii create acoustic reflections
- One discontinuity[4]:
  - Creates an "echo" chamber
- Multiple discontinuities:
  - Consistent with Bragg's scattering

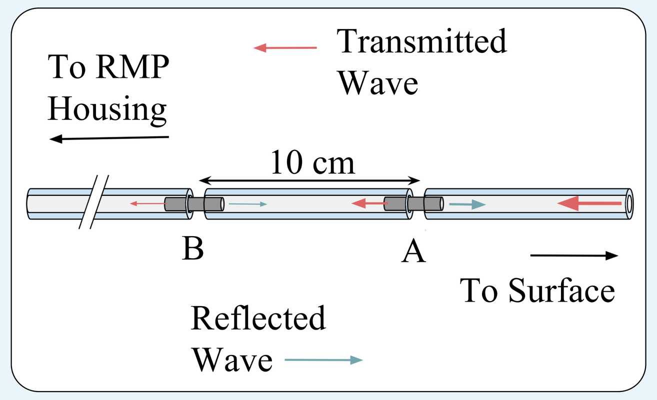


Figure 12: A tube containing multiple connectors.

Improvement: Gradually constrict radius; place connectors in non-periodic intervals

- TFs are sensitive to reference mic and speaker placement[5]
- Alignment in the axial direction is crucial for high coherence[6]

Improvement: Limit microphone and speaker deviations

Figure 13: New funnel holding reference mic and speaker at fixed relative positions.

Conclusion

- <u>Improved repeatability</u>: Reduced TF magnitude SD by **69%**
- <u>Improved quality</u>: Decreased tortuosity by **46%**; increased coherence by **4.6%** at < 300 Hz
- <u>Overall:</u> **Significant** additions to TF understanding. Provides a basis for further RMP improvements

References

[1] M. Awasthi, J. Rowlands, D. J. Moreau, and C. J. Doolan, "Two-step hybrid calibration of remote microphones," The Journal of the Acoustical Society of America, vol. 144, no. 5, pp. EL477–EL483, 2018, doi: 10.1121/1.5080462. AIP Publishing
[2] M. T. Q. S. da Silva, M. do R. Cardoso, C. M. P. Veronese, and W. Mazer, "Tortuosity: A brief review," Materials Today: Proceedings, vol. 58, pp. 1344–1349, 2022, doi: 10.1016/j.matpr.2022.02.228. ScienceDirect
[3] O. Moriaux, R. Zamponi, and C. Schram, "Semi-empirical calibration of remote microphone probes using Bayesian inference," Journal of Sound and Vibration, vol. 573, Art. no. 118197, 2024, doi: 10.1016/j.jsv.2023.118197. TU Delft Research Portal
[4] W. C. Chew, "Lecture 16: Waves in layered media," in Lectures on Electromagnetic Field Theory, Purdue University, West Lafayette, IN, USA, 2020. [Online]. Available: https://engineering.purdue.edu/wcchew/ece604f20/Lecture%20Notes/Lect16.pdf
[5] M. Adhikary, A. Pereira, M. Jacob, R. Blazquez-Navarro, and J. Thisse, "Experimental investigation of rotor shielding effects on axial flow fan using an interstage array," in Proc. 11th Convention of the European Acoustics Association (Forum Acusticum/Euronoise 2025), Málaga, Spain, 2025, pp. 1891–1898, doi: 10.61782/fa.2025.0377.
[6] Y. C. Küçükosman, C. Schram, and J. van Beeck, "A remote microphone technique for aeroacoustic measurements in large wind tunnels," Applied Acoustics, vol. 129, pp. 346–353, 2018, doi: 10.1016/j.apacoust.2017.08.018.